LAB 6 – Triple DES Algorithm

```
import java.util.*;
import javax.crypto.*;
import javax.xml.bind.DatatypeConverter;

class des3{
    static byte[] run(byte[] data, SecretKey key, int mode)
            throws Exception {
        Cipher c = Cipher.getInstance("DES/ECB/NoPadding");
        c.init(mode, key);
        return c.doFinal(data);
    }
    static void show(String text, byte[] data) {
        System.out.println(
            text + DatatypeConverter.printBase64Binary(data)
        );
    }

    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);
        KeyGenerator kg = KeyGenerator.getInstance("DES");

        SecretKey k1 = kg.generateKey();
        SecretKey k2 = kg.generateKey();
```

```
SecretKey k3 = kg.generateKey();

System.out.println("K1: " +
        DatatypeConverter.printHexBinary(k1.getEncoded()));
System.out.println("K2: " +
        DatatypeConverter.printHexBinary(k2.getEncoded()));
System.out.println("K3: " +
        DatatypeConverter.printHexBinary(k3.getEncoded()));

System.out.print("\nEnter exactly 8 characters: ");
String msg = sc.nextLine();

if (msg.getBytes("UTF-8").length != 8) {
   System.out.println("Message must be exactly 8 bytes.");
   return;
}

byte[] p = msg.getBytes("UTF-8");

System.out.println("\n--- Encryption ---");

byte[] a = run(p, k1, Cipher.ENCRYPT_MODE);
show("Encrypt using K1: ", a);
byte[] b = run(a, k2, Cipher.DECRYPT_MODE);
```

```
        show("Decrypt using K2: ", b);

        byte[] encrypted = run(b, k3, Cipher.ENCRYPT_MODE);
        show("Encrypt using K3: ", encrypted);

        System.out.println("\n--- Decryption ---");

        byte[] c = run(encrypted, k3, Cipher.DECRYPT_MODE);
        show("Decrypt using K3: ", c);

        byte[] d = run(c, k2, Cipher.ENCRYPT_MODE);
        show("Encrypt using K2: ", d);

        byte[] decrypted = run(d, k1, Cipher.DECRYPT_MODE);
        show("Decrypt using K1: ", decrypted);

        System.out.println(
            "\nFinal message: " +
            new String(decrypted, "UTF-8")
        );

        sc.close();
    }
}
```